Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

H_2S is bubbled into $0.2MNaCN$ solution which is $0.02M$ in each $[Cd(CN)_4]^{2-}$ and $[Ag(CN)_2]^-$. The H_2S produces $1 \times 10^{-9}M$ sulphide in the solution.
(Given, $K_{sp}Ag_2S = 1 \times 10^{-50}M^3; K_{sp}CdS = 7.1 \times 10^{-28}M^2$
$K_{inst}\ [Ag(CN)_2]^{-1} = 1 \times 10^{-20}M^2 \quad K_{inst}\ [Cd(CN)_4]^{-2} = 7.8 \times 10^{-18}M^4\Big)$
Identify the correct statement

(A) Ag_2S precipitates first from the solution

(B) CdS precipitates first from the solution

(C) None of them precipitate under the given conditions

(D) Ag_2S precipitates at a sulphide concentration of $1 \times 10^{-15}M$

Q.2

What is the maximum concentration of Ni^{+2} ions in water (at $25°C$) that is saturated with H_2S $(0.1M$ at $25°C)$ and maintained at pH 3 with HCl.K_{SP} of NiS is 3×10^{-21}.H_2S dissociates in two steps, each with an equilibrium constant -

$$H_2S \rightleftharpoons H^+ + HS^- \quad Ka_1 = 9.1 \times 10^{-8}$$
$$HS^- \rightleftharpoons H^+ + S^{-2} \quad Ka_2 = 1.1 \times 10^{-12}$$

(A) $3 \times 10^{-7}M$

(B) $4 \times 10^{-7}M$

(C) $5 \times 10^{-7}M$

(D) $6 \times 10^{-7}M$

Q.3

Solubility of $M(OH)_2$ is H_2O is $10^{-6}M$. Calculate its solubility in buffer solution of $pH = 8$.

(A) 4×10^{-12}

(B) 4×10^{-18}

(C) 4×10^{-7}

(D) 4×10^{-6}

Q.4

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K_a for cyanoacetic acid is 4×10^{-3}. What is the value of degree of hydrolysis of $0.4M$ sodium cyano acetate solution ?

(A) 4.5×10^{-6}

(B) 5.5×10^{-6}

(C) 2.5×10^{-6}

(D) 3.5×10^{-6}

Q.5

$100ml$ aqueous 0.1 molar $M(CN)_2(80\%$ ionized) solution is mixed with $100ml$ of 0.05 molar H_2SO_4 solution (80\% ionized) are mixed. What is the pH of the resultant solution at $25^{\circ}C$? $(K_b$ of $CN^- = 10^{-6})$

(A) 6

(B) 8

(C) 9

(D) 7

Q.6

The dissociation constants for aniline, acetic acid and water at $25^{\circ}C$ are $4 \times 10^{-10}, 2 \times 10^{-5}$ and 10^- 14 respectively. Calculate degree of hydrolysis of aniline acetate in a decinormal solution-

(A) 0.025

(B) 0.015

(C) 0.035

(D) 0.045

Q.7

Calculate the amount of $(NH_4)_2 SO_4$ in gm which must be added to $500ml$ of $0.2MNH_3$ to yield a solution of $pH = 9$, K_b for $NH_3 = 2 \times 10^{-5}$

(A) 3.248 mg

(B) 4.248 mg

(C) 1.320 mg

(D) 6.248 mg

Q.8

The K_a for formic acid and acetic acid are 2×10^{-4} and 2×10^{-5} respectively. Calculate the relative strength of acids with same molar concentration-

(A) $\sqrt{10}$

(B) $\sqrt{7}$

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(C) $\sqrt{8}$

(D) $\sqrt{5}$

Q.9

The pH of a $0.01\mathrm{M}$ solution of a monobasic acid is four. Which of the following statements about the acid is/are correct -

(A) When a little NaOH is added, it will form a buffer solution

(B) It is a weak acid

(C) It's sodium salt will be acidic

(D) It's sodium salt will be basic

Q.10

Calculate the pH of a $0.01\mathrm{M}$ solution of $\mathrm{H_2NCH_2CH_2NH_2}$, ethylenediamine (en).
Given: $\mathrm{K_{b_1}}$ and $\mathrm{K_{b_2}}$ values of ethylenediamine are $8.5 \times 10^{-5}\mathrm{M}$ and $7.1 \times 10^{-8}\mathrm{M}$ respectively.
Choose the correct options -

(A) The concentration of enH_2^{2+} in the solution is $9.2 \times 10^{-4}\mathrm{M}$

(B) pH of the solution is 10.97

(C) The concentration of enH_2^{2+} in the solution is $7.1 \times 10^{-8}\mathrm{M}$

(D) pH of the solution is 6.17

Q.11

Equal volumes of following solutions are mixed, in which case the pH of resulting solution will be average value of pH of two solutions.

(A) $\mathrm{pH}=2\ \mathrm{HCl}\ \&\ \ \mathrm{pH}=12\ \mathrm{NaOH}$

(B) $\mathrm{pH}=2\ \mathrm{HCl}\ \&\ \ \mathrm{pH}=4\ \mathrm{HCl}$

(C) $\mathrm{pH}=2\ \mathrm{HCN}\ \&\ \ \mathrm{pH}=12\ \mathrm{NaOH}\left(\mathrm{K_a}\ \text{of}\ \ \mathrm{HCN}=10^{-10}\right)$

(D) $\mathrm{pH}=5\ \mathrm{CH_3\ COOH}\ \&\ \ \mathrm{pH}=9\ \mathrm{NH_3(\ aq)},\ \mathrm{K_a\left(\ CH_3\ COOH\right)=\ K_b\left(\ NH_3(\ aq)\right)}$

Q.12

Paragraph:
A student was asked to determine the identity of an unknown acid that was liquid at room temperature $(20^{\circ}\mathrm{C})$. The student was told that the acid was one of those listed in Table . Characteristics of several acids:

Acid	Structure	Molecular weight (g/mole)	Melting point $(^{\circ}\mathrm{C})$	pKa
Propionic	$\mathrm{CH_3CH_2COOH}$	74.08	-21.5	4.88
Crotonic	$\mathrm{CH_3CH=CHCOOH}$	86.09	71.6	4.69
Butyric	$\mathrm{CH_3CH_2CH_2COOH}$	88.10	-7.9	4.82
Oxalic	$\mathrm{HOOCCOOH}$	90.04	101	3.14

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The student added $0.22\ \text{g}$ of the acid to $30.0\ \text{mL}$ of $\text{H}_2\text{O}(\ell)$. The student then titrated the solution with $0.10\text{M}\text{NaOH}(\text{aq})$ while monitoring the pH with a pH meter. The results are summarized in Figure.

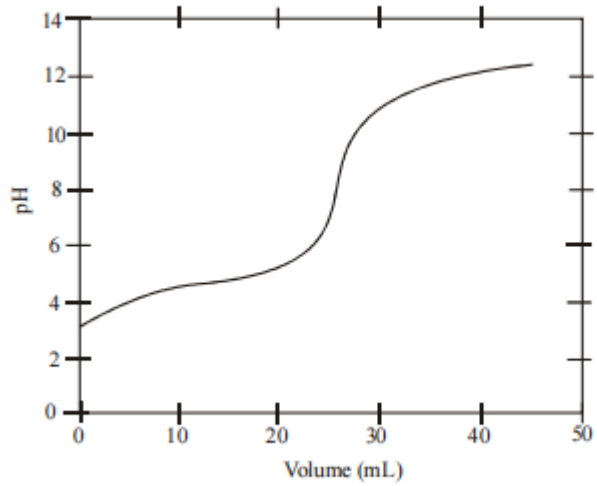


Figure : Titration of the acid with 0.1 M NaOH(aq)

Based on the titration curve, the student proposed that the unknown acid had 1 -COOH group and a molecular weight between 85 and 92 .

Question:

A comparison of which two compounds from Table best shows the effect of molecular weight alone on melting point?

(A) Propionic acid and crotonic acid

(B) Propionic acid and oxalic acid

(C) Propionic acid and butyric acid

(D) Butyric acid and crotonic acid

Q.13

Paragraph:

A student was asked to determine the identity of an unknown acid that was liquid at room temperature (20°C). The student was told that the acid was one of those listed in Table . Characteristics of several acids:

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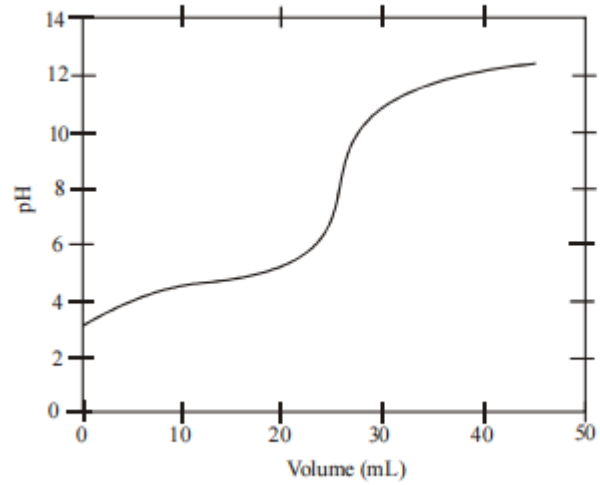


Figure : Titration of the acid with 0.1 M NaOH(aq)

Based on the titration curve, the student proposed that the unknown acid had 1 -COOH group and a molecular weight between 85 and 92 .

Question:

Before titrating with $NaOH(aq)$, what was the approximate $H_3O^+(aq)$ concentration of the solution containing the unknown acid

(A) $0.001 M$

(B) $0.01 M$

(C) $0.03 M$

(D) $0.3 M$

Q.14

Paragraph:

A student was asked to determine the identity of an unknown acid that was liquid at room temperature $(20^\circ C)$. The student was told that the acid was one of those listed in Table . Characteristics of several acids:

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The student added 0.22 g of the acid to 30.0 mL of $H_2O(\ell)$. The student then titrated the solution with

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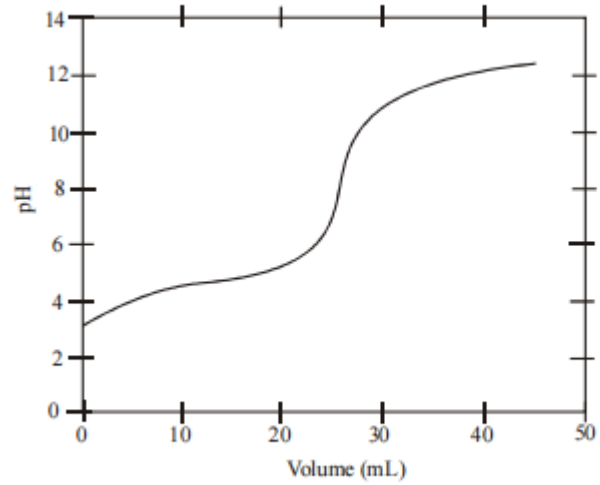


Figure : Titration of the acid with 0.1 M NaOH(aq)

Based on the titration curve, the student proposed that the unknown acid had 1 -COOH group and a molecular weight between 85 and 92 .

Question:

The student prepared a $0.1 M$ aqueous solution of crotonic acid and a $0.1 M$ aqueous solution of oxalic acid, then adjusted the pH of each to 4.7 by adding $NaOH$. Which solution has a lower freezing point?

(A) The crotonic acid solution, because it contains a lower molar concentration of solute particles

(B) The crotonic acid solution, because it contains a greater percent mass of solute

(C) The oxalic acid solution, because it contains a greater molar concentration of solute particles

(D) The oxalic acid solution, because it contains a smaller percent mass of solute

Q.15

Paragraph:

A certain quantity of lead chromate was accidentally spilled into a reservoir, and the city engineers would like to know to what extent drinking water was contaminated. The solubility product K_{sp} of lead chromate at $18^{\circ} C$ is 1.77×10^{-14}.

Question:

What is the solubility of lead chromate in pure water ?

(A) $1.33 \times 10^{-7} \mathrm{~mol/L}$

(B) $1.33 \times 10^{-2} \mathrm{~mol/L}$

(C) $1.33 \times 10^{-5} \mathrm{~mol/L}$

(D) $1.33 \times 10^{-4} \mathrm{~mol/L}$

Q.16

Paragraph:

A certain quantity of lead chromate was accidentally spilled into a reservoir, and the city engineers would like to know to what extent drinking water was contaminated. The solubility product K_{sp} of lead chromate at $18^{\circ} C$ is 1.77×10^{-14}.

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Question:

Some engineers believe that the lead in the water could be removed by treating it with potassium chromate (K_2CrO_4).

What is the solubility of lead chromate in $0.1\ mol/L$ of potassium chromate?

(A) 3.77×10^{-11}

(B) 2.77×10^{-11}

(C) 1.77×10^{-13}

(D) 1.77×10^{-15}

Q.17

Paragraph:

A certain quantity of lead chromate was accidentally spilled into a reservoir, and the city engineers would like to know to what extent drinking water was contaminated. The solubility product K_{sp} of lead chromate at $18^{\circ}C$ is 1.77×10^{-14}.

Question:

It was also believed that chromate ions could be removed from potable water by treating it with lead nitrate. What is the solubility of lead chromate in a $3.0 \times 10^{-7}\ mol/L$ solution of lead nitrate ?

(A) $0.505 \times 10^{-7}\ mol/L$

(B) $0.505 \times 10^{-5}\ mol/L$

(C) $0.505 \times 10^{-3}\ mol/L$

(D) $0.505 \times 10^{-9}\ mol/L$

Q.18

Math the column:

Column I	**Column II**
(A) The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4}. The pH of a 0.01 M solution of its sodium salt is	(p) 8
(B) A 40ml sample of an aqueous solution of the methylamine at 25°C is titrated with 0.15M HCl and the equivalence point is reached when 40ml of the acid have been added. The pH at the equivalence point is (Given : K_a for $CH_3NH_3^+$ is $\frac{4}{3} \times 10^{-11}$)	(q) 6
(C) Calculate the percentage protonation of 0.20M aqueous solution of methylamine CH_3NH_2. The K_b for CH_3NH_2 is 3.2×10^{-4}	(r) 4
(D) K_a of a weak acid (HA) is 10^{-5}. The pK_b of its conjugate base (B^-) is	(s) 9

(A) A-p; B-q ; C- r ; D-s;

(B) A-p; B-r ; C- s ; D-s;

(C) A-s; B-r ; C-s ; D-s;

(D) A-s; B-p ; C- p ; D-s;

Q.19

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Calculate the p^H at which the following conversion (reaction) will be at equilibrium in basic medium
$I_2(s) \rightarrow I^-(aq) + I_3^-(aq)$
When the equilibrium concentration at 300k are $[\text{I}^-] = 0.10\text{M}$ and $\left[\text{IO}_3^-\right] = 0.10\text{M}$ { Given
$\rightarrow \Delta G_f^0\left(I^-, aq\right) = -50\ \text{kJ/mol}\Big\}$
$\Delta G_f^0 = \left[1O_3^-, aq\right] = -123.5\ \text{kJ/mol}$
$\Delta G_f^0 = (\text{H}_2\text{O}, l) = -233\ \text{kJ/mol}$
$\Delta G_f^0 = \left(\text{OH}^-, aq\right) = -150\ \text{kJ/mol}$
R (Gas constant) $= \frac{25}{3} J/\text{mol} - K$
$\log e = 2.3\}$

Q.20

K_a for HCN is 5×10^{-10} at 25°C. For maintaining a constant $\text{pH} = 9$, the volume of 5MKCN solution required to be added to 10ml of 2MHCN solution is
[Given $\log_{10} 5 = 0.69$, antilog $(-0.3010) = 0.5$]

Answers & Solutions

Q.1 Answer:

CdS precipitates first from the solution

Solution:

From question

$\text{Ag(CN)}_2^- \rightleftharpoons \text{Ag}^+ + 2\text{CN}^- \quad \text{K}_\text{inst} = 1 \times 10^{-20}\text{M}^2$
$\text{Cd(CN)}_4^{-2} \rightleftharpoons \text{Cd}^{+2} + 4\text{CN}^- \quad \text{K}_\text{inst} = 7.8 \times 10^{-18}\text{M}^4$
$\text{Cd(CN)}_4^{-2} \rightleftharpoons \text{Cd}^{+2} + 4\text{CN}^-$
$\therefore \quad \left[\text{Ag}^+\right] = \frac{\text{K}_\text{inst} \times [\text{Ag}_\text{inst}\,(\text{CN})_2]}{[\text{CN}^{-2}]^2} = \frac{1\times10^{-20}\times0.02}{(0.2)^2} = 5 \times 10^{-21}\text{M}$
Similarly, $\left[\text{Cd}^{+2}\right] = \frac{7.8\times10^{-18}\times0.02}{(0.2)^4}\text{M} = 9.75 \times 10^{-17}\text{M}$
$\left[\text{S}^{-2}\right]$ **needed to precipitate** $\left[\text{Ag}^+\right]$
$= \frac{\text{K}_\text{sp}\times[\text{Ag}_2\,\text{S}]}{[\text{Ag}^+]^2} = \frac{1\times10^{-50}}{(5\times10^{-21})^2} = 4 \times 10^{-10}\text{M}$
$\left[\text{S}^{-2}\right]$ **needed to precipitate** $\left[\text{Cd}^{+2}\right]$
$= \frac{\text{K}_{sp}\times[\text{CdS}]}{[\text{Cd}^{+2}]} = 7.203 \times 10^{-12}\text{M}$
Since $\left[\text{S}^{-2}\right]$ **needed to precipitate** Cd^{+2} **ion is smaller than that of** Ag^+ **ion. Therefore** Cd^{+2} **will precipitate first.**

Q.2 Answer:

$3 \times 10^{-7}\text{M}$

Solution:

The overall dissociation of $\text{H}_2\,\text{S}$ **will be the product of** K_{a_1} **and** K_{a_2}**.**
$\text{K}_\text{a} = \text{K}_{\text{a}_1} \times \text{K}_{\text{a}_2} = \left(9.1 \times 10^{-8}\right) \times \left(1.1 \times 10^{-12}\right) = 1 \times 10^{-19}$
Saturated $\text{H}_2\,\text{S}$ **is approximately** 0.1M **and dissociation of** $\text{H}_2\,\text{S}$ **is very slight**
So, $\left[\text{H}^+\right]^2 \left[\text{S}^{-2}\right] = 1 \times 10^{-19} \times 10^{-1} = 10^{-20}$

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Since, pH is 3, $[H^+] = 10^{-3}M$

$[S^{-2}] = \frac{10^{-20}}{[H^+]^2} = \frac{10^{-20}}{10^{-6}} = 10^{-14}M$

Since NiS will precipitate, if the S.P. is exceeded, the highest value which [Ni+2 concentration can have is:

$[Ni^{+2}] = \frac{3\times10^{-21}}{10^{-14}} = 3 \times 10^{-7}M$

Q.3 Answer:

4×10^{-6}

Solution:

$M(OH)_2 \rightleftharpoons M_s^{+2} + 2OH_{2\,S}^-$

$K_{sp} = [M^{+2}] [OH^-]^2$

$= (S) \quad (2S)^2 = 4S^3$

$= 4 \left(10^{-6}\right)^3 = 4 \times 10^{-18}$

In buffer $pH = 8$

$pOH = 6$

$[OH^-] = 10^{-6}$

$M(OH)_2 \rightleftharpoons M^{-2} + 2OH^-$

$K_{sp} = [M^{+2}] [OH^-]^2$

$4 \times 10^{-18} = [M^{+2}] [10^{-6}]^2$

$\frac{4\times10^{-18}}{10^{-12}} = [M^{+2}]$

$[M^{2+}] = 4 \times 10^{-6}$

Q.4 Answer:

2.5×10^{-6}

Solution:

$\therefore \quad$ **Moles of Cl in the final solution** $= 0.04 \ \mathrm{K}) \cdot K_h = \frac{K_W}{K_a} = \frac{10^{-14}}{4\times10^{-3}} = 0.25 \times 10^{-11} \ \mathrm{h} = \sqrt{\frac{K_h}{c}} = \sqrt{\frac{0.25\times10^{-11}}{0.4}} = 2.5 \times 10^{-6}$

Q.5 Answer:

8

Solution:

$100\mathrm{ml}0.1\mathrm{M}M(CN)_2$ **contains 20 meq of** $M(CN)_2$

$100\mathrm{ml}0.05\mathrm{M} \quad H_2SO_4$ **contains 10 meq of** H_2SO_4

80% **ionized** $M(CN)_2$ **gives** $16\mathrm{meq}$ **of** CN^-

80% **ionized** H_2SO_4 **gives 8 meq of** H^+

$\therefore$ **A buffer of** $HCN + CN^-$ **with equal no of equivalents is formed.**

$\therefore \quad pH = pKa = -\log 10^{-s} = 8$

$\because K_a = \frac{10^{-14}}{K_b}$ at $25^\circ C$

Q.6 Answer:

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0.035

Solution:

Aniline $^{+}$ **+Acetate** $^{-}$ $+ \mathrm{H_2O} \rightleftharpoons$ **Aniline + Acetic acid**

Before hydrolysis 1 $\quad 1 \quad 0 \quad 0$

$$\text{Let concen, salt be C mole litre}^{-1}$$
$$\mathrm{K_h} = \frac{\mathrm{Ch} \cdot \mathrm{Ch}}{\mathrm{C}(1-\mathrm{h}) \cdot \mathrm{C}(1-\mathrm{h})} = \frac{\mathrm{h}^2}{(1-\mathrm{h})^2}$$
$$\frac{\mathrm{h}}{1-\mathrm{h}} = \sqrt{\frac{\mathrm{K_W}}{\mathrm{K_a} \cdot \mathrm{K_b}}} = \sqrt{\frac{10^{-14}}{2 \times 10^{-6} \times 4 \times 1}}$$
$$\% \text{ hydrolysis } = 54.95\%$$
$$\mathrm{h} = 0.035$$

Q.7 Answer:

1.320 mg

Solution:

$$\mathrm{pOH} = -\log \mathrm{K_b} + \log \frac{\left[\mathrm{NH_4^+}\right]}{[\mathrm{NH_4OH}]}$$

Let 'a' millimoles of $\mathrm{NH_4^+}$ **are added to a solution**

having milli moles of $\mathrm{NH_4OH} = 500 \times 0.2 = 100$

$\therefore \quad [\mathrm{NH_4}^+] = [\text{salt }] = \frac{\mathrm{a}}{500}$ and $[\mathrm{NH_4OH}] = [\text{ Base }]$

Given $\mathrm{K_b}$ for $\mathrm{NH_4OH} = 2 \times 10^{-5}$ and $\mathrm{pH} = 9$

$\therefore \quad 5 = -\log 2 \times 10^{-5} + \log \frac{\mathrm{a}/500}{100/500}$

$\therefore \quad \mathrm{a} = 200 \text{mili moles } = 0.2 \text{ mol}$

moles of $(\mathrm{NH_4})_2\,\mathrm{SO_4}$ added $= \frac{\mathrm{a}}{2} = 0.1$ mol

$\therefore \mathrm{W}_{(\mathrm{NH_4})_2\mathrm{SO_4}} = 0.1 \times 132 = 1.32$

Q.8 Answer:

$\sqrt{10}$

Solution:

Relative strength of weak acids $= \sqrt{\left(\frac{\mathrm{K_{a_1}}}{\mathrm{K_{a_2}}} \times \frac{\mathrm{C_1}}{\mathrm{C_2}}\right)}$

$\therefore \quad$ **Relative strength** $= \sqrt{\frac{\mathrm{K_{a_1}}}{\mathrm{K_{a_2}}}} (\because \mathrm{C_1} = \mathrm{C_2}) = \sqrt{\left(\frac{2 \times 10^{-4}}{2 \times 10^{-5}}\right)}$

Relative strength for HCOOH **to** $\mathrm{CH_3COOH} = \sqrt{10} : 1$

Q.9 Answer:

When a little NaOH is added, it will form a buffer solution

Solution:

Concentration of monobasic acid $= 0.01\mathrm{M}$

$\mathrm{pH} = 4$

If the acid is completely ionised the pH **of the acid would be** $\mathrm{pH} = -\log 0.01 = -\log 10^{-2} = 2$

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So it is a weak acid. The sodium salt of a weak acid when dissolved the anions will be hydrolysed giving rise to OH^- ion concentration. The solution will be basic. So statement (C) is incorrect.

Q.10 Answer:

pH of the solution is 10.97

Solution:

Let the base etylenediamine is represented by en.

$$en + H_2O \rightleftharpoons enH^+ + OH^-; \quad K_{b_1} = \frac{[enH^+][OH^-]}{[en]} = 8.5 \times 10^{-5}$$
$$enH^+ + H_2O \rightleftharpoons enH_2^{2+} + OH^-; K_{b_2} = \frac{[enH_2^{2+}][OH^-]}{[enH^+]} = 7.1 \times 10^{-8}$$

Assuming that OH^- coming from enH^+ is insignificant and the total $[OH^-]$ in the solution is approximately equal to $[OH^-]$ coming from en only.

$$\therefore \quad K_{b_1} = \frac{x^2}{0.01} = 8.5 \times 10^{-5}$$
$$\text{so, } x = 9.2 \times 10^{-4} M = [OH^-] = [enH^+]$$
$$pOH = 3.03 \text{ and } pH = 10.97$$
$$K_{b_2} = \frac{(9.2 \times 10^{-4})[enH_2^{2+}]}{9.2 \times 10^{-4}} = 7.1 \times 10^{-8}$$
$$\text{so } [enH_2^{2+}] = 7.1 \times 10^{-8} M$$

Q.11 Answer:

$pH = 2$ HCl & $pH = 12$ NaOH

Solution:

(a) $pH = 2HCl$ **so** $[HCl] = 10^{-2}$

$pH = 12NaOH$ **so** $[NaOH] = 10^{-2}$

Hence pH **of mixture** $= 7$ **(neutral)**

(d) $pH = 5CH_3COOH$ **and** $pH = 9NH_3$ **(aq), both must be of equal concentrations as** $pK_a = pK_b$.

Hence $pH = pOH$ **only if concentrations are equal so on mixing equal volume we will get** CH_3COONH_4 **salt solution and its** pH **is given by**

$pH = \frac{1}{2}\{pK_w + pK_a - pK_b\} = 7$

Q.12 Answer:

Propionic acid and butyric acid

Solution:

According to the data in Table, both structure and molecular weight (i.e., molar mass) affect the melting point of a compound. In order to assess the effect of molecular weight or mass alone, any other effects such as obvious structural differences must be minimized. This is best done by comparing two compounds that are structurally similar. Because the structures of propionic acid and butyric acid (Answer c) differ by only a CH_2 group, they best show that melting point increases with molar mass. All of the other answer

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choices compare two compounds that differ significantly in structure. Therefore, the melting points of these compounds include both molar mass and structural effects. Thus, answer choice *c* is the best answer.

Q.13 Answer:

$0.001\mathrm{M}$

Solution:

Figure shows the pH of the solution to be about 3 before any $\mathrm{NaOH(aq)}$ is added.

$\mathrm{pH} = -\log\left[\mathrm{H_3O^+}\right]; 3 = -\log\left[\mathrm{H_3O^+}\right]$

$\left[\mathrm{H_3O^+}\right] = 10^{-3}\mathrm{M} = 0.001\mathrm{M}$

Q.14 Answer:

The oxalic acid solution, because it contains a greater molar concentration of solute particles

Solution:

The freezing point depression of an aqueous solution is a colligative property (i.e., it depends on the number of solute particles in a given volume of water.) Given two solutions, the one with the greater number of solute particles per liter of solution freezes at the lower temperature. Answer C is the only answer that relates a larger number of solute particles directly to a lower freezing point. Oxalic acid is diprotic and ionizes in accord with the pKa values in Table 1 to a greater extent than does crotonic acid. Subsequently, oxalic acid requires more NaOH than does crotonic acid to reach a pH of 4.7, and oxalic acid produces a larger number of particles in solution. Thus, answer choice *c* is the best answer.

Q.15 Answer:

$1.33 \times 10^{-7}\ \mathrm{mol/L}$

Solution:

(a) The solubility product $\mathrm{K_{sp}}$ of lead chromate at $18^{\circ}\mathrm{C}$ is 1.77×10^{-14}.

The dissolution of lead chromate is represented by the following equation:

$\mathrm{PbCrO_4(\ s) \rightarrow Pb^{2+} + CrO_4^{2-}}$

Therefore, in pure water, the solubility of lead chromate is equal to the concentration of either the $\mathrm{Pb^{2+}}$ or the $\mathrm{CrO_4^{2-}}$ in a saturated salt solution.

Therefore, $\mathrm{Pb^{2+} = CrO_4^{2-} = SPbCrO_4 = \sqrt{K_{sp}} = 1.33 \times 10^{-7}\ mol/L}$

(b) In such a solution, the solubility of lead chromate is no longer equal to the concentration of chromate ions since a large part of these ions are provided by potassium chromate. However, its solubility is equal to the concentration of lead ions all provided by lead chromate.

Therefore, $\mathrm{S_{PbCrO_4} = S^{2+}_{Pb} = \frac{K_{sppbro\,4}}{[CrO_4{}^{2-}]} = \frac{1.77\times10^{-14}}{0.1} = 1.77 \times 10^{-13}}$

Note: There are two sources of chromate ions in solution. Therefore, some of the lead chromate will precipitate from solution.

(c) $\mathrm{SPbCrO_4 = CrO_4^{2-} = \frac{K_{spPbCrO_4}}{3.00\times10^{-7}+\left[CrO_4^{2-}\right]} = \frac{1.77\times10^{-14}}{3.00\times10^{-7}+\left[CrO_4^{2-}\right]} = 0.505 \times 10^{-7}\ mol/L}$

Q.16 Answer:

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1.77×10^{-13}

Solution:

(a) The solubility product K_{sp} of lead chromate at $18^{\circ}C$ is 1.77×10^{-14}.

The dissolution of lead chromate is represented by the following equation:

$PbCrO_4(s) \rightarrow Pb^{2+} + CrO_4^{2-}$

Therefore, in pure water, the solubility of lead chromate is equal to the concentration of either the Pb^{2+} or the CrO_4^{2-} in a saturated salt solution.

Therefore, $Pb^{2+} = CrO_4^{2-} = SPbCrO_4 = \sqrt{K_{sp}} = 1.33 \times 10^{-7}\ mol/L$

(b) In such a solution, the solubility of lead chromate is no longer equal to the concentration of chromate ions since a large part of these ions are provided by potassium chromate. However, its solubility is equal to the concentration of lead ions all provided by lead chromate.

Therefore, $S_{PbCrO_4} = S_{Pb}^{2+} = \frac{K_{sppbro_4}}{[CrO_4{}^{2-}]} = \frac{1.77 \times 10^{-14}}{0.1} = 1.77 \times 10^{-13}$

Note: There are two sources of chromate ions in solution. Therefore, some of the lead chromate will precipitate from solution.

(c) $SPbCrO_4 = CrO_4^{2-} = \frac{K_{spPbCrO_4}}{3.00 \times 10^{-7} + [CrO_4^{2-}]} = \frac{1.77 \times 10^{-14}}{3.00 \times 10^{-7} + [CrO_4^{2-}]} = 0.505 \times 10^{-7}\ mol/L$

Q.17 Answer:

$0.505 \times 10^{-7}\ mol/L$

Solution:

(a) The solubility product K_{sp} of lead chromate at $18^{\circ}C$ is 1.77×10^{-14}.

The dissolution of lead chromate is represented by the following equation:

$PbCrO_4(s) \rightarrow Pb^{2+} + CrO_4^{2-}$

Therefore, in pure water, the solubility of lead chromate is equal to the concentration of either the Pb^{2+} or the CrO_4^{2-} in a saturated salt solution.

Therefore, $Pb^{2+} = CrO_4^{2-} = SPbCrO_4 = \sqrt{K_{sp}} = 1.33 \times 10^{-7}\ mol/L$

(b) In such a solution, the solubility of lead chromate is no longer equal to the concentration of chromate ions since a large part of these ions are provided by potassium chromate. However, its solubility is equal to the concentration of lead ions all provided by lead chromate.

Therefore, $S_{PbCrO_4} = S_{Pb}^{2+} = \frac{K_{sppbro_4}}{[CrO_4{}^{2-}]} = \frac{1.77 \times 10^{-14}}{0.1} = 1.77 \times 10^{-13}$

Note: There are two sources of chromate ions in solution. Therefore, some of the lead chromate will precipitate from solution.

(c) $SPbCrO_4 = CrO_4^{2-} = \frac{K_{spPbCrO_4}}{3.00 \times 10^{-7} + [CrO_4^{2-}]} = \frac{1.77 \times 10^{-14}}{3.00 \times 10^{-7} + [CrO_4^{2-}]} = 0.505 \times 10^{-7}\ mol/L$

Q.18 Answer:

A-p; B-q ; C- r ; D-s;

Solution:

((A) - p, (B) - q, (C) - r, (D) - s)

(A) pH of sodium salt of weak acid $= \frac{1}{2}(pK_w + pK_a + \log C) = \frac{1}{2}(14 + 4 - 2) = 8$

(B) $CH_3NH_2 + HCl \longrightarrow CH_3NH_3 + Cl^-$

At eq. $-\quad -\quad 40 \times 0.15 = 6$ **m mole point**

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$CH_3\overset{\oplus}{N}H_3$ **undergo hydrolysis while** Cl^- **remain unhydrolysed**

$CH_3NH_3^+ + H_2O \rightleftharpoons CH_3NH_2 + H_3O^+$

$\frac{6}{80}(1-h) \qquad \frac{6}{80}\,h$

$h = \sqrt{\frac{K_a \times 80}{6}}\,h$

$[H_3O^+] = \frac{6}{80}\,h = \sqrt{K_a \times \frac{6}{80}} = \sqrt{\frac{4}{3} \times 10^{-11} \times \frac{6}{80}} = \sqrt{10^{-12}}$

$[H_3O^+] = 10^{-6}M$

$\therefore \quad pH = 6.0$

(C) $CH_3NH_2(aq) + H_2O(\ell) \rightleftharpoons CH_3NH_3^+ + OH^-(aq)$

$0.20 - x$

$\therefore \quad K_b = \frac{x^2}{0.20-x}$, neglecting x in the denominator

$\therefore \quad x = \sqrt{0.20\,(3.2 \times 10^{-4})} = 8 \times 10^{-3}$

$\therefore \quad \text{Percentageprotonation} = CH_3NH_2 = \frac{x}{0.20} \times 100 = \frac{8 \times 10^{-3}}{0.20} \times 100 = 4\%$

(D) Given $K_a = 10^{-5}, pK_a = -\log K_a = -\log 10^{-5} = 5$

Now, $pK_a + pK_b = 14$ (at $25^\circ C$)

$pK_b = 14 - pK_a = 14 - 5 = 9$

Q.19 Answer:

—

Solution:

Balanced equation is $3I_2(s) + 6OH^- \rightleftharpoons 5I^- + IO_3^- + 3H_2O(l)$

$\Delta G^0 = 5 \times (-50) + (-123.5) + 3 \times (-233) \times -0 - 6 \times (-150) = -172.5\ \text{kJ/mol}$

Now

$\Delta G^0 = -RT \ln k$

$\Rightarrow -172.5 = -\frac{25}{3} \times 300 \times 2.3 \times 10^{-3} \log 3$

$\Rightarrow \log k = 30$

$\Rightarrow 10^{30} = \frac{[I^-]^5 \times [IO_3^-]}{[OH^-]^6} = \frac{10^{-5} \times 10^{-1}}{[OH^-]^6}$

$\Rightarrow [OH^-] = 10^{-6}$

$P^{OH} = 6$

$P^H = 8$

Q.20 Answer:

—

Solution:

$pH = -\log 5 \times 10^{-10} + \log\left(\frac{5\,V}{20}\right)$, **where** V **is the volume of** KCN **solution in** ml.

$\therefore \quad V = 2ml$

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